

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH EXAMINATIONS - NOVEMBER 2025

Analog VLSI Design [ECE_5153]

Marks: 50

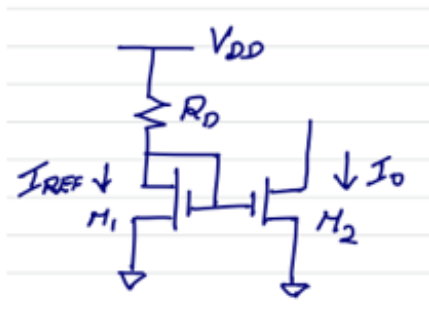
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) For the below NMOS current mirror comment on the sensitivity to VDD. (4)

A)



Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Consider VDD to be a constant voltage and the noise on VDD as a small signal applied on VDD and derive equation for sensitivity to VDD. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) What would you do to make the sensitivity of I_{REF} to VDD less. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Design a circuit that can self-bias itself (BMR) to generate a constant reference current. Design BMR to give a constant bias current of 25uA (4)

A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) What are the two stable operating conditions in which the BMR can stably operate and are both the states desirable? Explain. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Design a circuit to nudge the BMR to start in case reference current becomes zero. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) How do you reduce the sensitivity of the reference current to VDD in BMR? (10)

- 4) Discuss the different configurations of a single stage amplifier. (5)
How are the G_m and R_{out} varying for each configuration?

A)

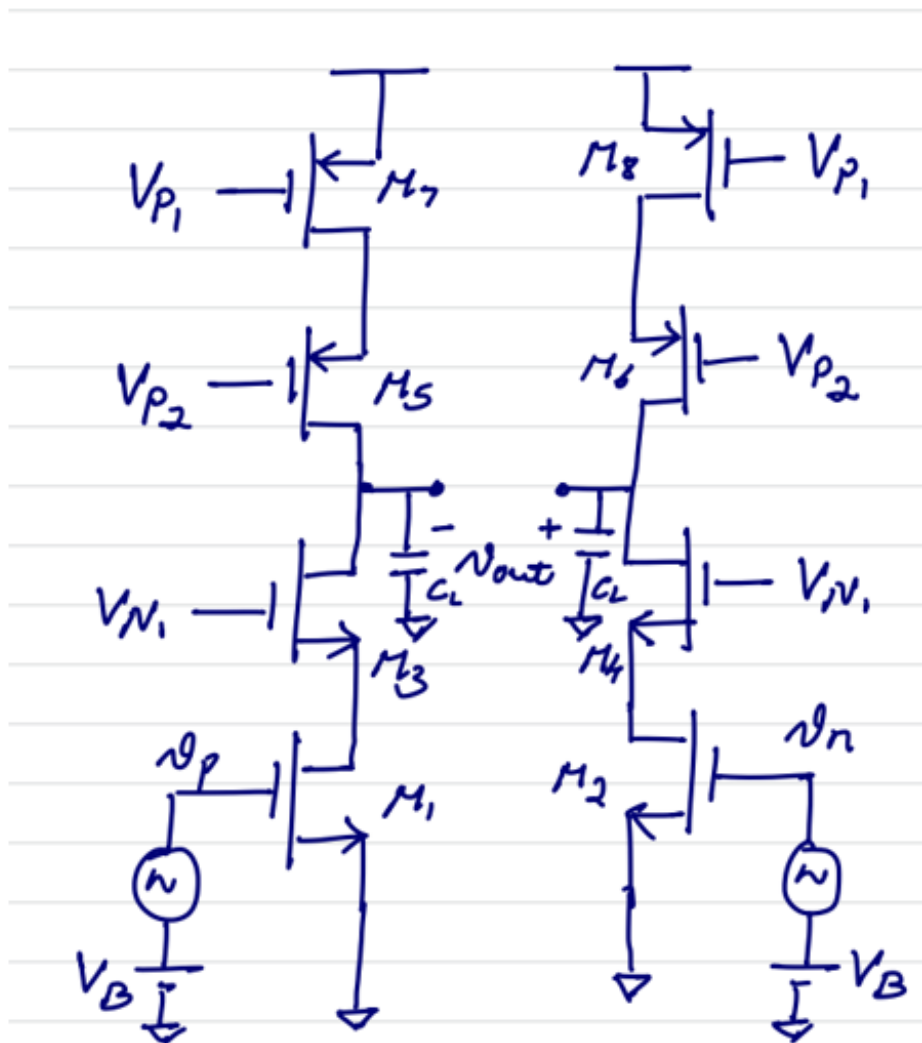
- B) What is the maximum gain achievable and what are its implications on bandwidth? (5)

- 5) Design Telescopic Differential amplifier with maximum load capacitance of 1pF to have the following. (5)

A)

Gain > 80db. BW > 9kHz. Power < 120 μ W. Minimum Swing between 200 mV to 1.6mV.

- B) Use BMR designed in Question 2 to bias the MOSFETs in saturation. (5)



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Event Name	Subject
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ECE_5153:Analog VLSI

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG ECE_5153:Analog VLSI Design

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	7	25(50%)
4	Analysis	4	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		11	50 (100%)